

# A Mathematical Explanation of How Parameters Are Balanced in Zenless Zone Zero's Damage Calculations

## (Part 2)

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### **Introduction**

Anomaly damage is calculated using a weighted average of the anomaly accumulation value, current ATK, damage bonuses, Anomaly Mastery, and other factors. While the anomaly accumulation value is set for each agent's skill, it cannot be viewed on the in-game status screen.

Therefore, this article introduces a method for estimating the anomaly accumulation value for each skill and the enemy's anomaly accumulation cap by using the formula for Abloom damage, which takes the anomaly accumulation value into account.

### **Content of this time**

1. Damage Calculation Method for “Abloom” (Without Considering Anomaly Accumulation)
2. Damage Calculation Method for “Abloom” (With Considering Anomaly Accumulation)
3. Method for Estimating Anomaly Accumulation Values from Skills
4. Method for Estimating the Enemy's Anomaly Accumulation Cap

\* Please note that the purpose here is simply to provide specific examples related to “AOD” and “Sanguine Sweeper”

## **Damage Calculation Method for Abloom (Without Considering Anomaly Accumulation)**

### **Factors Contributing to Abloom Damage**

1. ATK - Agent Base Attack, Attack Buff
2. Damage Bonus – Initial Value 100 and the total of Disk 5 Main, etc.
3. Anomaly Proficiency – Initial values for each agent and the total of sub-stats, etc.
4. Defense Modifier – Enemy Defense, Defense Reduction, Penetration Value, Penetration Rate, etc.
5. Element Weakness – Adjustments based on the enemy's weakness element, etc.
6. Stun DMG Multiplier – Stun-time Correction
7. Anomaly Damage Bonus – Bonus adjustment specifically for anomaly damage
8. Level Adjustment – At Agent Level 60, there are 2
9. Anomaly Damage Multiplier – Fixed values set for each attribute
10. Abloom Multiplier – Multiplier per agent
11. Abloom Damage Bonus – Bonus adjustment specific to Abloom Damage
12. Separate multiplication – Items other than those listed above

### **Example of ATK Calculation for Aria (AOD Team)**

| Category                   | Value | Breakdown                                           |
|----------------------------|-------|-----------------------------------------------------|
| Initial Value              | 1576  | Base ATK : 788<br>Core Skill : 75<br>Aria Sign: 713 |
| Pre-battle Multiplier Buff | 1.63  | Substat: 63                                         |
| Battle Multiplier Buff     | 1.1   | Chinatsu Sign: 10                                   |
| Pre-battle Buff            | 335   | Disc 2 Main: 316<br>Sub-stat actual value: 19       |
| Battle Buffs               | 1100  | Chinatsu Core Passive : 1050<br>Ether Veil: 50      |

$$\text{ATK} = \text{Floor}(1576 \times 1.63 + 335) \times 1.1 + 1100 = 4293.3$$

Here, for any positive real number  $x$ ,  $\text{Floor}(x)$  represents the integer part of  $x$ .

### Example of Damage Bonus Calculation for Aria (AOD Team)

| Category        | Value | Breakdown                                                                                                                        |
|-----------------|-------|----------------------------------------------------------------------------------------------------------------------------------|
| Initial Value   | 100   | Common Value: 100                                                                                                                |
| Pre-battle Buff | 30    | Disc 5 Main: 30                                                                                                                  |
| Battle Buffs    | 143   | Chinatsu Sign: 25<br>Nangong Core Skill: 25<br>Nangong Sign: 30<br>Aria Sign: 20<br>Phaethon Melody: 25<br>Moonlight Lullaby: 18 |

$$\text{Damage Bonus} = 100 + 30 + 143 = 273$$

### Example of Anomaly Proficiency Calculation for Aria (AOD Team)

| Category        | Value | Breakdown                                                   |
|-----------------|-------|-------------------------------------------------------------|
| Initial Value   | 116   | Aria initial value: 116                                     |
| Pre-battle Buff | 248   | Disc 4 Main: 92<br>Substat: 126<br>Chaos Jazz 2pc: 30       |
| Battle Buffs    | 225   | Aria Sign: 90<br>Aria Core Skill: 90<br>Phaethon Melody: 45 |

$$\text{Anomaly Proficiency} = 116 + 248 + 225 = 589$$

### Formula for calculating defense adjustment

- $D_{enemy}$  : Enemy Defense. Determined by the enemy's type and level.
- $r_{pen}$  : PEN Ratio
- $v_{pen}$  : Flat PEN
- $r_{down}$  : DEF Down, Ignore DEF
- L: Agent level adjustment value. When the agent level is 60, L = 794

Enemy Defense After Debuff E =  $D_{enemy}(1 - r_{pen})(1 - r_{down}) - v_{pen}$

$$\text{Defense Modifier} = \frac{L}{E+L}$$

### Example of Defense Modifier Calculation for Aria (AOD Team)

| Category                        | Value | Breakdown                                  |
|---------------------------------|-------|--------------------------------------------|
| Enemy Defense                   | 84.6  | Thanatos Training Center<br>Level 10: 84.6 |
| PEN Ratio                       | 0     |                                            |
| Flat PEN                        | 18    | Substat: 18                                |
| DEF Down                        | 0     |                                            |
| Agent Level Adjustment<br>Value | 794   |                                            |

Enemy Defense After Debuff =  $84.6 \times (1-0)(1-0) - 18 = 66.6$

$$\text{Defense Modifier} = \frac{794}{66.6 + 794} = \frac{794}{860.6}$$

### Example of Element Weakness Calculation for Aria (AOD Team)

| Category             | Value | Breakdown                    |
|----------------------|-------|------------------------------|
| Initial Value        | 100   | Common initial value: 100    |
| Enemy-specific bonus | 20    | Thanatos (Weak to Ether): 20 |
| Debuff               | 0     |                              |

Element Weakness =  $100 + 20 + 0 = 120$

### Example of Stun DMG Multiplier Calculation for Aria (AOD Team)

| Category      | Value | Breakdown                                                         |
|---------------|-------|-------------------------------------------------------------------|
| Initial Value | 200   | Thanatos Training Center Initial Value: 200                       |
| Debuff        | 60    | Chinatsu Additional Ability: 30<br>Nangong Additional Ability: 30 |

$$\text{Stun DMG Multiplier} = 200 + 60 = 260$$

The “Anomaly Damage Bonus” referred to here refers to what is added as the same item as Aria’s sign “Attribute Anomaly DMG +10%.”

### Example of Anomaly Damage Bonus Calculation for Aria (AOD Team)

| Category      | Value | Breakdown                 |
|---------------|-------|---------------------------|
| Initial Value | 100   | Common initial value: 100 |
| Buff & Debuff | 10    | Aria Sign: 10             |

$$\text{Anomaly Damage Bonus} = 100 + 10 = 110$$

\*Also, there is an additional ability for Yuzuha in the same category.

For the purposes of this discussion, we will treat level adjustments and anomaly damage multipliers as fixed values, so we will omit the details.

### Examples of Level Adjustments and Anomaly Damage Multipliers for Aria (AOD Team)

| Category                  | Value | Notes                             |
|---------------------------|-------|-----------------------------------|
| Level Adjustment          | 200   | Aria Level 60                     |
| Anomaly Damage Multiplier | 62.5  | Fixed Corrupted Damage Multiplier |

The method for calculating the “Abloom” damage multiplier varies depending on the character's skill. For Aria, the “Abloom” damage multiplier is determined by the base multiplier based on her Core Skill level and her Anomaly Mastery value.

### Example of Abloom Damage Multiplier for Aria (AOD Team)

| Category              | Value | Notes                                                       |
|-----------------------|-------|-------------------------------------------------------------|
| Base Multiplier       | 27.5  | Core Skill F<br>While under the Corrupted<br>Anomaly effect |
| Anomaly Mastery Value | 253   | Core Skill F<br>2pc Phaethon Melody<br>Disc 6: Main (AM)    |

$$\text{Abloom Damage Multiplier} = \frac{27.5 \times 253}{10} = 695.75$$

The “Abloom Damage Bonus” referred to here refers to modifiers that are added together as a single value, such as the 50% buff granted by Aria's core passive ability when a stun occurs, and the 30% buff from the “Metamorph” DA buff.

### Example of Abloom Damage Bonus with Aria (AOD Team)

| Category        | Value | Breakdown         |
|-----------------|-------|-------------------|
| Base Multiplier | 100   |                   |
| Buff & Debuff   | 50    | Aria Core Passive |

$$\text{Abloom Damage Bonus} = 100 + 50 = 150$$

Items that do not fall under any of the categories listed above are grouped together as a separate multiplier. In this case, this refers to the “Anomaly DMG taken is increased by 15%” listed in the enemy details for “Sanguine Sweeper” in Deadly Assault.

The product of all the values calculated so far is as follows. Except for ATK and Defense Modifier, all values are divided by 100 to convert them from percentages back to ratios.

$$4293.3 \times \frac{273}{100} \times \frac{589}{100} \times \frac{794}{860.6} \times \frac{120}{100} \times \frac{260}{100} \times \frac{110}{100} \times \frac{200}{100} \times \frac{62.5}{100} \times \frac{695.75}{100} \times \frac{150}{100}$$

$$= 2851609.844...$$

We will refer to the value obtained by calculating the product of these parameters as “Calculated Damage.” On the other hand, we will refer to the damage actually displayed in the game as “Displayed Damage.” In this paper, we assume that the displayed damage is the calculated damage rounded up to the nearest integer.

### **Assumption: The Relationship between Calculated Damage and Displayed Damage**

The Calculated Damage  $d$  and Displayed Damage  $D$  satisfy the following conditions.

$$D = \text{Ceil}(d)$$

$$\text{Example: } 2851610 = \text{Ceil}(2851609.844...)$$

Each agent's ability has a base anomaly accumulation value set for it. This does not depend on the agent's level or skill level.

### Formula for calculating anomaly accumulation value

$q$  : Basic anomaly accumulation value set per skill

$P_{anom}$  : Anomaly threshold value

$B_{acc}$  : Anomaly accumulation efficiency

$R_{enemy}$  : Enemy attribute resistance

$$t = q \times \frac{P_{anom}}{100} \times \frac{B_{acc}}{100} \times \frac{R_{enemy}}{100}$$

If multiple attacks are made before a anomaly occurs, the ATK is calculated using a weighted average based on the anomaly accumulation value dealt by each attack, rather than a simple average.

### Weighted average of ATK

Let  $X_i$  be the ATK of the  $i$  attack ( $i = 1, \dots, n$ ) and  $t_i$  be the anomaly accumulation value. In this case, the ATK  $\bar{X}$  used to calculate anomaly damage is given by the following formula.

$$T = t_1 + t_2 + \dots + t_n$$

$$\bar{X} = \frac{t_1 X_1 + t_2 X_2 + \dots + t_n X_n}{t_1 + t_2 + \dots + t_n}$$

Here, T represents the upper limit of the enemy's anomaly accumulation.

For some items, the weighted average of the values at the time of anomaly accumulation is used, while for others, the values at the time the damage occurred are used.



| Items to include in the snapshot                                                   | Items calculated in real time                                                                       |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 1. ATK<br>2. Damage Bonus<br>3. Anomaly Proficiency<br>4. Flat PEN<br>5. PEN Ratio | 1. Stun DMG Multiplier<br>2. Anomaly Damage Bonus<br>3. Abloom Multiplier<br>4. Abloom Damage Bonus |

\* Items not listed above were not included in this investigation regarding whether they are captured in a snapshot.

## Method for Approximating Anomaly Skill Accumulation Values

Let  $t$  be the target anomaly accumulation value for the desired skill (including all multiplicative factors such as Anomaly Mastery), and let  $T$  be the enemy's maximum anomaly accumulation value. If you first hit the enemy with a skill while your ATK is  $X$ , then buff your ATK by  $\Delta x$ , and finally accumulate the remaining anomaly accumulation value, the weighted average of the ATK is

$$\bar{X} = \frac{tX + (T - t)(X + \Delta x)}{T} = X + \left(1 - \frac{t}{T}\right)\Delta x$$

Assuming there are no buffs other than ATK, and letting the product of the other factors be a constant  $C$ , the calculated damage  $d$  is

$$d = C\bar{X} = C \left\{ X + \left(1 - \frac{t}{T}\right)\Delta x \right\}$$

Assuming that the calculated damage and the displayed damage satisfy  $D = \text{Ceil}(d)$ , we have  $D - 1 < d \leq D$ . Solving for  $\frac{t}{T}$  within this range yields the following:

**Range of the relative value  $\frac{t}{T}$**

$$\frac{C(X + \Delta x) - D}{C\Delta x} \leq \frac{t}{T} < \frac{C(X + \Delta x) - (D - 1)}{C\Delta x}$$

In this way, it is possible to determine the range of relative values; however, to find the absolute value  $t$ , the value of the enemy's anomaly accumulation maximum  $T$  is necessary.

Since the anomaly accumulation caps for regular enemies and like are calculated as shown below, we will use these values. Note that the anomaly accumulation caps for Deadly Assault bosses often differ significantly from the values shown in the image.

| 异常积蓄阈值                             |      |      |      |      |      |      |      |      |      |      |
|------------------------------------|------|------|------|------|------|------|------|------|------|------|
| ! 数值来源：公测实测                        |      |      |      |      |      |      |      |      |      |      |
| 属性异常<br>触发次数                       | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9+   |
| 普通敌人                               | 600  | 612  | 624  | 636  | 648  | 660  | 673  | 686  | 699  | 712  |
| 普通敌人<br>(物理)                       | 720  | 734  | 748  | 762  | 777  | 792  | 807  | 823  | 839  | 855  |
| 精英敌人                               | 2250 | 2295 | 2340 | 2386 | 2433 | 2481 | 2530 | 2580 | 2631 | 2683 |
| 精英敌人<br>(物理)                       | 2700 | 2754 | 2809 | 2865 | 2922 | 2980 | 3039 | 3099 | 3160 | 3223 |
| 首领敌人                               | 3000 | 3060 | 3121 | 3183 | 3246 | 3310 | 3376 | 3443 | 3511 | 3581 |
| 首领敌人<br>(物理)                       | 3600 | 3672 | 3745 | 3819 | 3895 | 3972 | 4051 | 4132 | 4214 | 4298 |
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Anomaly Accumulation Threshold

! Data Source: Public Beta Testing Results

| Number of<br>Attribute Anomaly<br>Triggers | 0    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9+   |
|--------------------------------------------|------|------|------|------|------|------|------|------|------|------|
| Normal Enemy                               | 600  | 612  | 624  | 636  | 648  | 660  | 673  | 686  | 699  | 712  |
| Normal Enemy<br>(Physical)                 | 720  | 734  | 748  | 762  | 777  | 792  | 807  | 823  | 839  | 855  |
| Elite Enemy                                | 2250 | 2295 | 2340 | 2386 | 2433 | 2481 | 2530 | 2580 | 2631 | 2683 |
| Elite Enemy<br>(Physical)                  | 2700 | 2754 | 2809 | 2865 | 2922 | 2980 | 3039 | 3099 | 3160 | 3223 |
| Boss Enemy                                 | 3000 | 3060 | 3121 | 3183 | 3246 | 3310 | 3376 | 3443 | 3511 | 3581 |
| Boss Enemy<br>(Physical)                   | 3600 | 3672 | 3745 | 3819 | 3895 | 3972 | 4051 | 4132 | 4214 | 4298 |

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## Range of Absolute Values

For the range obtained in the previous slide, multiplying the entire inequality by  $T$  gives the range of the skill's anomaly accumulation value  $t$ .

$$\frac{C(X + \Delta x) - D}{C\Delta x} \cdot T \leq t < \frac{C(X + \Delta x) - (D - 1)}{C\Delta x} \cdot T$$

Also, the width of the interval becomes as follows.

$$\Delta t = T \cdot \frac{C(X + \Delta x) - (D - 1)}{C\Delta x} - T \cdot \frac{C(X + \Delta x) - D}{C\Delta x} = \frac{T}{C\Delta x}$$

In other words, the larger the product  $C$  (factors other than ATK) and the ATK buff amount  $\Delta x$ , the narrower the interval width becomes, allowing for a more accurate approximation.

The enemy used for testing is Training Ground Thanatos. Additionally, to maximize the ATK buff value  $\Delta x$ , the investigation is conducted using a team consisting of Aria, Chinatsu, and Caesar.

### Example of ATK Buffs for Aria (Chinatsu–Caesar Team Composition)

| Category                         | Value | Breakdown / Notes                                                                |
|----------------------------------|-------|----------------------------------------------------------------------------------|
| Enemy Anomaly Accumulation Limit | 2250  | Training Ground Thanatos                                                         |
| ATK Buff                         | 2100  | Chinatsu Core Passive: 1050<br>Caesar Additional Ability: 1000<br>Ether Veil: 50 |

Enemy Anomaly Accumulation Limit  $T = 2250$

Attack Buff  $\Delta x = 2100$

Since only ATK must be buffed in this test, equipment or effects that also increase Damage Bonus—such as Yuzuha, Chinatsu's signature W-Engine, or Moonlight 4-piece—cannot be used.

To keep Damage Bonus and Anomaly Mastery fixed, the Drive Discs and W-Engine are adjusted from the previous example. The resulting changed parameters and values are shown below.

### Example Parameters for Aria (Chinatsu–Caesar Team Composition)

| Category            | Value | Breakdown                                     |
|---------------------|-------|-----------------------------------------------|
| Pre-Battle ATK      | 2951  | Nangong signature W-Engine, substats, etc.    |
| Damage Bonus        | 140   | Drive Disc Slot 5 Main Stat, Aria 2-piece Set |
| Anomaly Mastery     | 427   | Chaos 2-piece Set, Substat: 99, etc.          |
| Stun DMG Multiplier | 230   | Chinatsu Core Passive                         |

Determine the base value of Aria's EX Special Attack: "Fall Into Delusion". Since the displayed damage  $D$  is 745,881, the range of  $t$  becomes as follows:

$$\frac{C(X+\Delta x) - D}{C\Delta x} \cdot T = 1387.4507,$$

$$\frac{C(X+\Delta x) - (D - 1)}{C\Delta x} \cdot T = 1387.456096.$$

$$1387.4507 \leq t < 1387.456096$$

Furthermore, by dividing by the Anomaly Proficiency value 2.53 and the Attribute Resistance factor 1.2, the skill's base value  $q$  falls within the following range:

$$456.999572 \leq q < 457.0013491$$

The remaining difference is a matter of preference, so here Aria's EX Special Attack: "Fall Into Delusion". is assigned a base value of 457.00. Using the same method, the base values of other skills were also calculated to the second decimal place; part of the results are shown below.

#### **Aria's Skill Base Anomaly Accumulation Values**

| Skill Name                                                                   | Base Value |
|------------------------------------------------------------------------------|------------|
| Basic Attack: Sweet Melody – Hit 1                                           | 22.80      |
| Basic Attack: Sweet Melody – Hit 2                                           | 46.57      |
| Basic Attack: Sweet Melody – Hit 3                                           | 58.04      |
| Basic Attack: Sweet Melody – Hit 4                                           | 134.00     |
| Dash Attack: Silky-Smooth Combo                                              | 20.81      |
| Basic Attack: Perfect Pitch – Charge Level Three Attack                      | 130.00     |
| Basic Attack: Perfect Pitch – Charge Level Three Attack (Coordinated Attack) | 149.50     |
| EX Special Attack: Instantly Hooked                                          | 486.20     |
| Chain Attack: Dream Collab                                                   | 542.46     |

Below are some of Nangong's base values, calculated using a similar method.

#### **Nangong's Skill Base Anomaly Accumulation Values**

| Skill Name                                                                               | Base Value |
|------------------------------------------------------------------------------------------|------------|
| Basic Attack: Shooting Star Step – Hit 1                                                 | 59.33      |
| Basic Attack: Shooting Star Step – Hit 2                                                 | 62.15      |
| Basic Attack: Shooting Star Step – Hit 3                                                 | 129.07     |
| Basic Attack: Adorable Explosive Impact – Charge Level One Attack                        | 40.00      |
| Basic Attack: Adorable Explosive Impact – Charge Level Two Attack                        | 191.63     |
| Basic Attack: Adorable Explosive Impact – Charge Level Three Attack (Coordinated Attack) | 373.36     |
| EX Special Attack: The Unbearable Weight of Love                                         | 463.27     |

## Method for Approximating the Enemy's Anomaly Accumulation Limit

Using the previously determined skill anomaly accumulation values, we can estimate the anomaly accumulation limit of the Deadly Assault boss "**Sanguine Sweeper**". Starting from the formula obtained earlier:

$$\frac{C(X + \Delta x) - D}{C\Delta x} \leq \frac{t}{T} < \frac{C(X + \Delta x) - (D - 1)}{C\Delta x}$$

Solving this equation for  $T$  gives:

$$\frac{tC\Delta x}{C(X + \Delta x) - (D - 1)} < T \leq \frac{tC\Delta x}{C(X + \Delta x) - D}$$

Thus, we obtain the range for  $T$ .

However, even if the same Caesar–Chinatsu team composition is used as before, it is not possible to obtain a value that appears highly accurate.

|        |             |
|--------|-------------|
| スキル基礎値 | 22.8        |
| 異常掌握   | 208         |
| 蓄積効率   | 100         |
| 耐性     | 120         |
| 基礎値端点① | 3509.154518 |
| 基礎値端点② | 3505.04193  |

|                                   |             |
|-----------------------------------|-------------|
| Skill Base Value                  | 22.8        |
| Anomaly Proficiency               | 208         |
| Accumulation Efficiency           | 100         |
| Resistance                        | 120         |
| Endpoint ① of Estimated Threshold | 3509.154518 |
| Endpoint ② of Estimated Threshold | 3505.04193  |

Therefore, we also apply buffs to Damage Bonus, Anomaly Mastery, and other factors in order to increase the amount of information available.

Let the increases provided by buffs be  $\Delta x$ ,  $\Delta y$ , and  $\Delta z$  respectively. Then the weighted averages of ATK, Damage Bonus, and Anomaly Mastery, denoted by  $\bar{X}$ ,  $\bar{Y}$ , and  $\bar{Z}$ , are:

### ATK Weighted Average

$$\bar{X} = \frac{tX + (T - t)(X + \Delta x)}{T} = X + \left(1 - \frac{t}{T}\right) \Delta x$$

### Damage Bonus Weighted Average

$$\bar{Y} = \frac{tY + (T - t)(Y + \Delta y)}{T} = Y + \left(1 - \frac{t}{T}\right) \Delta y$$

### Anomaly Mastery Weighted Average

$$\bar{Z} = \frac{tZ + (T - t)(Z + \Delta z)}{T} = Z + \left(1 - \frac{t}{T}\right) \Delta z$$

Thus, they can be expressed as above.

The calculated damage is:

Substituting the weighted-average values from the previous slide:

$$d = \frac{C}{10000} \left\{ X + \left(1 - \frac{t}{T}\right) \Delta x \right\} \left\{ Y + \left(1 - \frac{t}{T}\right) \Delta y \right\} \left\{ Z + \left(1 - \frac{t}{T}\right) \Delta z \right\}$$

Thus. Here, let

$$C' = \frac{C}{10000}, \quad w = \frac{t}{T}$$

Then, by defining these variables,

$$d = C'X + (1 - w)\Delta xY + (1 - w)\Delta yZ + (1 - w)\Delta z$$

We obtain the above expression.



Treat the calculated damage  $d$  as a function of the relative value  $w$ , denoted  $d = d(w)$ . The domain is defined as  $0 \leq w \leq 1$ . Define functions representing the difference between the calculated damage and the displayed damage bounds:

$$F(w) = d(w) - D$$

$$G(w) = d(w) - (D - 1)$$

We find approximate values of  $w$  for which these functions become zero on the interval  $0 \leq w \leq 1$  using the **bisection method**. Under the assumption below, both  $F$  and  $G$  are monotonically decreasing on  $0 \leq w \leq 1$ , so each has exactly one solution in that interval.

### **Assumption: Range of Displayed Damage**

Assume that the displayed damage is sufficiently larger than the calculated damage before buffs, and sufficiently smaller than the calculated damage after buffs. In other words,  $d(1) \ll D - 1 < D \ll d(0)$  is assumed to hold.

### **Interval Update Using the Bisection Method**

Let the initial interval be:

$$w_L^{(0)} = 0, w_R^{(0)} = 1$$

Define the midpoint at the  $k$ -th iteration as:

$$w_M^{(k)} = \frac{w_L^{(k)} + w_R^{(k)}}{2}$$

Then determine the interval endpoints for iteration  $k + 1$  according to the following cases:

Case 1

$$F(w_L^{(k)})F(w_M^{(k)}) \leq 0 \begin{cases} w_L^{(k+1)} = w_L^{(k)} \\ w_R^{(k+1)} = w_M^{(k)} \end{cases}$$

If the sign changes between the left endpoint and the midpoint

Keep the left half of the interval.

Case 2

$$F(w_L^{(k)})F(w_M^{(k)}) > 0 \begin{cases} w_L^{(k+1)} = w_M^{(k)} \\ w_R^{(k+1)} = w_R^{(k)} \end{cases}$$

If the sign does not change between the left endpoint and the midpoint

Keep the right half of the interval.

After performing the bisection method  $n$  times, the interval width becomes:

$$w_R^{(n)} - w_L^{(n)} = \frac{w_R^{(0)} - w_L^{(0)}}{2^n} = \frac{1}{2^n}$$

Therefore, define the approximate solution as:

$$w = \frac{w_L^{(n)} + w_R^{(n)}}{2}$$

Then, the maximum approximation error is:

$$\left| w - \frac{w_L^{(n)} + w_R^{(n)}}{2} \right| \leq \frac{w_R^{(n)} - w_L^{(n)}}{2} = \frac{1}{2^{n+1}}$$

Thus, the error is bounded by:

$$\frac{1}{2^{n+1}}$$

For a sufficiently large  $n$ , let the approximate solutions obtained from  $F(w) = 0$  and  $G(w) = 0$  be denoted by  $w_F$  and  $w_G$ , respectively.

Then, we expect that

$$d(w_G) < d(w) \leq d(w_F)$$

(While this is guaranteed for the exact mathematical solutions, additional verification would be required for approximate solutions. For simplicity, we assume it also holds for the approximations here.)

Since the calculated damage  $d$  is monotonically decreasing on the interval  $0 \leq w \leq 1$ ,

$$w_F \leq w < w_G$$

follows.

Therefore, the enemy's anomaly accumulation threshold  $T$  is bounded by

$$\frac{t}{w_G} < T \leq \frac{t}{w_F}.$$

Setting the number of iterations to

$$n = 70,$$

and using the base anomaly value of Aria's **“Dash Attack: Silky-Smooth Combo”**

$$t = 20.81,$$

the displayed damage

$$D = 4,048,310,$$

and the factors summarized on the following slide, the estimated range of the anomaly accumulation threshold  $T$  for **“Sanguine Sweeper”** is:

$$3506.237931 \dots < T \leq 3506.279388 \dots$$

Although this estimate is not extremely precise, an error of this magnitude has little impact on damage calculations.

For simplicity, we take the average value and set

$$T = 3506.25.$$

Furthermore, it is assumed that the anomaly accumulation threshold increases by **2% for each Corruption stack**.

### Values Used

| Parameter                                | Value                      | Used in Formula                |
|------------------------------------------|----------------------------|--------------------------------|
| ATK                                      | 2903 $\rightarrow$ 6028.75 | $X = 2903, \Delta x = 3125.75$ |
| Damage Bonus                             | 130 $\rightarrow$ 249      | $Y = 130, \Delta y = 119$      |
| Anomaly Mastery                          | 544 $\rightarrow$ 649      | $Z = 544, \Delta z = 105$      |
| Defense Multiplier                       | 0.6339828                  | Included in $C'$               |
| Defense Multiplier x<br>Level Correction | 125                        | Included in $C'$               |
| Abloom Multiplier                        | 1461.075                   | Included in $C'$               |
| Elemental Resistance                     | 120                        | Included in $C'$               |
| Stun DMG Multiplier                      | 180                        | Included in $C'$               |
| Separate Multiplicative<br>Modifier      | 172.5                      | Included in $C'$               |